

Supplementary Material

S1. Pristine vs. Irradiated TEM microstructural characterization

The deposited films exhibited a dense microstructure with no observable void formation associated with the PVD process. Figure S1 presents a cross-sectional TEM image of the irradiated Cu-24Cr-5Ta (at.%) thin film after exposure to 25 DPA. The upper region corresponds to the irradiation-damaged area, while the lower region remains undamaged. The SRIM-calculated profiles of injected interstitials and generated vacancies are shown on the left side of the figure for reference. On the right side of the figure, an EDS analysis of the top and bottom regions of the lamella is presented, showing that Cr and Ta are predominantly co-localized in the precipitates, more visible in the undamaged region.

A clear irradiation-induced microstructural evolution can be observed, primarily driven by irradiation-enhanced precipitate dissolution. In the pristine condition, the microstructure consists of elongated grains approximately ~ 150 nm in length and ~ 30 – 40 nm in width. In contrast, the irradiated region exhibits a significantly altered morphology, where the precipitates are largely dissolved and adopt a more equiaxed shape rather than the original elongated morphology. Additionally, the matrix displays a more homogeneous solid-solution-like elemental distribution after irradiation.

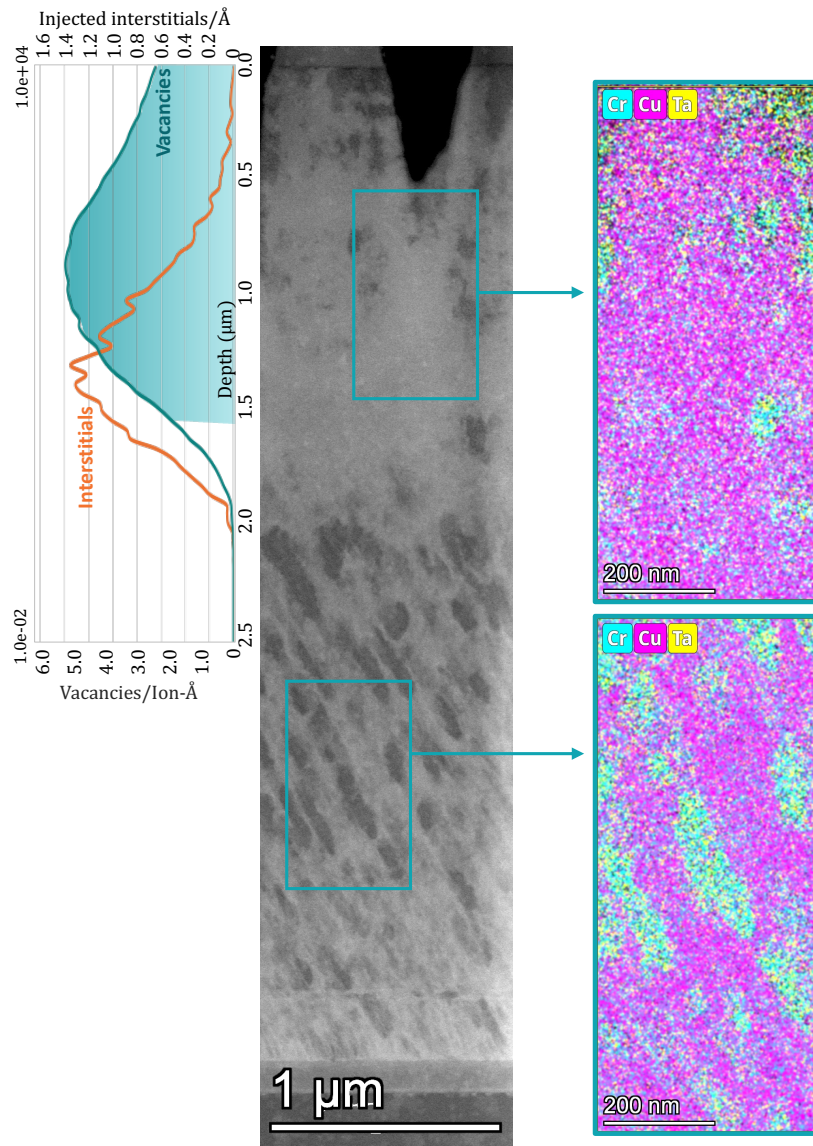


Figure S1: TEM cross section of the irradiated Cu-24Cr-5Ta (at.%) thin film after exposure to 25 DPA. Upper region corresponds to the irradiation-damaged area, while the lower region remains undamaged. On the right, an EDS analysis of the top and bottom regions of the lamella is presented, showing that Cr and Ta are predominantly co-localized in the precipitates.

S2. Point defect rate equation adaptation consideration

The equilibrium concentration of each of these species is determined by recombination/annihilation and the creation of new point defects during consecutive radiation cascades. Then, the point defect balance equations in the presence of successive radiation cascades are:

$$\begin{aligned}\frac{dC_v}{dt} &= K_0(1 - \epsilon_r)(1 - \epsilon_v) - K_{iv}C_iC_v - K_{vs}C_vC_s + L_v \\ \frac{dC_i}{dt} &= K_0(1 - \epsilon_r)(1 - \epsilon_i) - K_{iv}C_iC_v - K_{is}C_iC_s\end{aligned}\tag{S1}$$

where the subscripts i , v and s denote interstitials, vacancies, and sinks, respectively, $C[\#/m^3]$ is the concentration, $K_0[DPA/s]$ is the production rate, $K_{xy}[m^3/s]$ is the recombination rate coefficient for components denoted by the subscripts x and y . $L_v[\#/m^3s]$ is the production of thermal vacancies from sinks, ϵ_r is the on-cascade vacancy-interstitial recombination rate and ϵ_i or ϵ_v is the fraction of interstitials/vacancies that will migrate to a pre-existing cluster of the same species [8]. Then, sink strengths are defined by:

$$K_{vs} = 4\pi r_{vs}D_v\tag{S2}$$

$$K_{is} = 4\pi r_{is}D_i\tag{S3}$$

$$K_{iv} = 4\pi r_{iv}(D_i + D_v)\tag{S4}$$

but as $D_i \gg D_v$,

$$K_{iv} \approx 4\pi r_{iv}D_i\tag{S5}$$

In nanocrystalline materials, as is the case of the samples presented here, the cascade recombination rate between interstitials and vacancies is smaller than for non-nanocrystalline materials, as mobile interstitials will find a grain boundary far more rapidly than a vacancy [46], so 0.1 has been selected for ϵ_r , implying that around 10% of the vacancies will recombine with interstitials. On the other hand, at low temperatures, vacancy clusters increase the energy of the system, being an unstable defect, so they tend to recombine or eject vacancies. Additionally, saturated grain boundaries will eject interstitials from them that will recombine with pre-existing defects, like stacking-fault

tetrahedral (SFT), voids, etc, before some vacancies have the opportunity to find them [47, 48]. Considering all of this, at room temperature irradiations the fraction of vacancies that contribute to pre-existing vacancy clusters is negligible, making $\epsilon_v = 0$. Finally, it is important to mention that at low temperatures ($T \leq 0.5T_m$, where T_m is the melting temperature) the production of thermal vacancies from sink sources is not significant [8], then $L_v = 0$. Considering all these conditions, the expression for vacancy evolution in equation (S1) can be simplified to:

$$\frac{dC_v}{dt} = 0.9K_0 - K_{iv}C_iC_v - K_{vs}C_vC_s \quad (\text{S6})$$

The time frame in which the SAW frequency evolves and stabilizes when the beam is turned on and off in our experiments (see Figure 3.a), is similar to the time frame in which the vacancy/interstitial concentrations evolve, on the order of minutes [36] for the experimental conditions presented here, therefore we investigated the relation of the point defect evolution formulas with the SAW frequency. Previous computational simulations from E. Martinez et al. [36], employing standard rate theory (SRT) and atomistic kinetic Monte Carlo (akMC), have shown that the time frame in which the period of mutual recombination occurs during the equilibration process of point defects at the beginning of the irradiation, is in the range between 0.0005 - 200 seconds for an irradiation performed at 300K and using a dose rate of 10^{-3}DPA/s (see Figure S2), which matches the relaxation times that we observe in our experiments in similar conditions.

As seen in Figure 3, the evolution of the SAW frequency shows a different behavior during the beam onset and offset, implying that different but related mechanisms are taking place in the material by the beam fluctuation. For this reason, we evaluate point defect concentration evolution for these two cases independently. When the beam is turned on, point defects in the material are created by the impinging ion beam (K_0), some of them will recombine between them (K_{iv}), and others will annihilate at grain boundaries (K_{vs}). Then, equation S6 describes what happens with the interstitial concentration by the beam onset for a high sink material irradiated at low temperatures (RT) [8]. By equating this expression, the vacancy concentration in the regime prior when point defect equilibrium is reached, where mutual recombination is greater than interstitial recombination at sinks, would be:

$$C_v = \left(\frac{0.9K_0K_{vs}C_s t}{K_{iv}} \right)^{\frac{1}{2}} = \left(\frac{0.9K_0 4\pi r_{is} D_i C_s t}{4\pi r_{iv} (D_v + D_i)} \right)^{\frac{1}{2}} \approx (0.9K_0 C_s t)^{\frac{1}{2}} \quad (\text{S7})$$

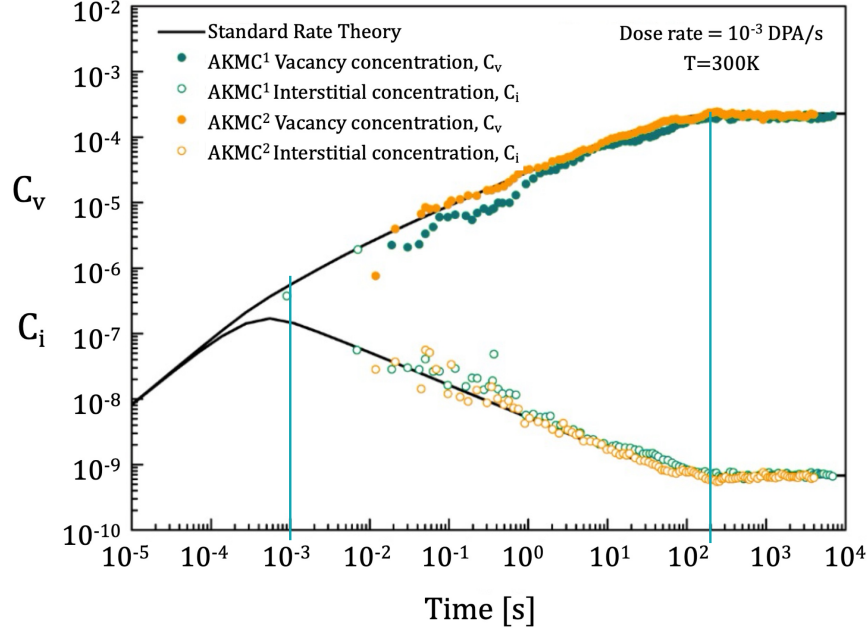


Figure S2: Point defect evolution for FeNi at 300K and 10-3 DPA/s, obtained from standard rate theory and atomistic kinetic Monte Carlo (AKMC) simulations using a random sink distribution and an annihilation rate of sinks. Plot adapted from [36].

Later, when the beam is turned off, there will be no creation of point defects, so $K_0 = 0$, and also all interstitials will have migrated to the grain boundary, therefore there will be no recombination between vacancy and interstitials, $K_{iv} = 0$. After applying these conditions to equation S6, the following expression can be derived:

$$\ln \frac{C_v}{C_{v_0}} = -K_{vs}C_s(t - t_0) \quad (\text{S8})$$

$$C_v = C_{v_0}e^{-K_{vs}C_s t} \quad (\text{S9})$$

S3. Calculation of the contribution of interstitials vs. vacancies

The following calculation demonstrated that the interstitial term on equation 16 of the main text can be considered of negligible influence respect the

vacancy term, due to the difference of three orders of magnitude between them.

$$\frac{dV_{\text{void}}}{dt} = 4\pi R\Omega [D_v (C_v - C_v^V) - D_i C_i] \quad (\text{S10})$$

At $t = 0.0001$, $C_v^V \approx 0$. For copper, the following values are used:

$$D_v = 3.9 \times 10^{-7} \text{ cm}^2\text{s}^{-1} [45]$$

$$C_v \approx 2 \times 10^{-4} \text{ cm}^{-3} [36]$$

$$D_i = 3.19 \times 10^{-5} \text{ cm}^2\text{s}^{-1} [45]$$

$$C_i \approx 6 \times 10^{-10} \text{ cm}^{-3} [36]$$

Thus,

$$D_v C_v \approx 7.8 \times 10^{-11}$$

$$D_i C_i \approx 1.9 \times 10^{-14}$$

Therefore,

$$D_v C_v \gg D_i C_i \quad (\text{S11})$$

S4. Model considerations for different conditions

As discussed during the development of this model, several assumptions were introduced to enable an accurate representation of the material behavior. Consequently, for different microstructures or experimental conditions, new formulations may be required to capture specific material responses.

One of the key assumptions concerns the irradiation temperature. Since the experiments were conducted at $T \leq 0.5T_m$, it was assumed that the production of thermal vacancies from sinks is negligible. At these temperatures, both vacancies and sinks lack sufficient mobility and energy to enable vacancy emission. However, for higher irradiation temperatures ($T \geq 0.5T_m$), enhanced diffusivity must be considered. Under these conditions, parameters such as L_v and ϵ_v would change due to increased vacancy mobility, which in turn promotes faster defect clustering and the formation of larger voids.

For this reason, it is important to assess how an increasing void population affects the elastic properties of the material, particularly when extending the model to higher void volume fractions. Garner & Wolfer ([49], pages 58-69) conducted a study on the evolution of elastic properties as a function of void volume fraction, with the aim of evaluating the impact of void swelling

on physical properties. In their work, they derived expressions describing the effective variation of properties such as bulk modulus, Poisson’s ratio, Young’s modulus, and thermal conductivity. These calculations are based on the Hashin [50] model, itself derived from the Eshelby formalism [51], which evaluates the effect of an ellipsoidal inclusion, treated here as a void, on the effective material response. Garner & Wolfer exhibit the evolution of Poisson’s ratio, Young’s modulus, and thermal conductivity as a function of increasing void fraction. It is shown that a void volume fraction on the order of 1% represents a reasonable estimate for materials prone to void formation, where vacancies cluster into significant cavities. Within this range, and up to approximately 0.1 volume fraction, changes in the elastic modulus remain approximately linear. In contrast, thermal conductivity exhibits greater variability, with limited agreement across different models and, in some cases, clear deviations from linear behavior. Nevertheless, these results suggest that for void volume fractions up to about 1%, the assumptions and equations used in this model remain valid.

This analysis also supports the capability of TGS to provide reliable estimates of vacancy concentration evolution. Variations in vacancy concentration lead to measurable changes in elastic properties, which, through the dependence of surface acoustic wave velocity on these properties, as described in [27, 24], result in detectable shifts in SAW velocity. These shifts are directly reflected in the TGS signal, enabling indirect characterization of defect evolution.